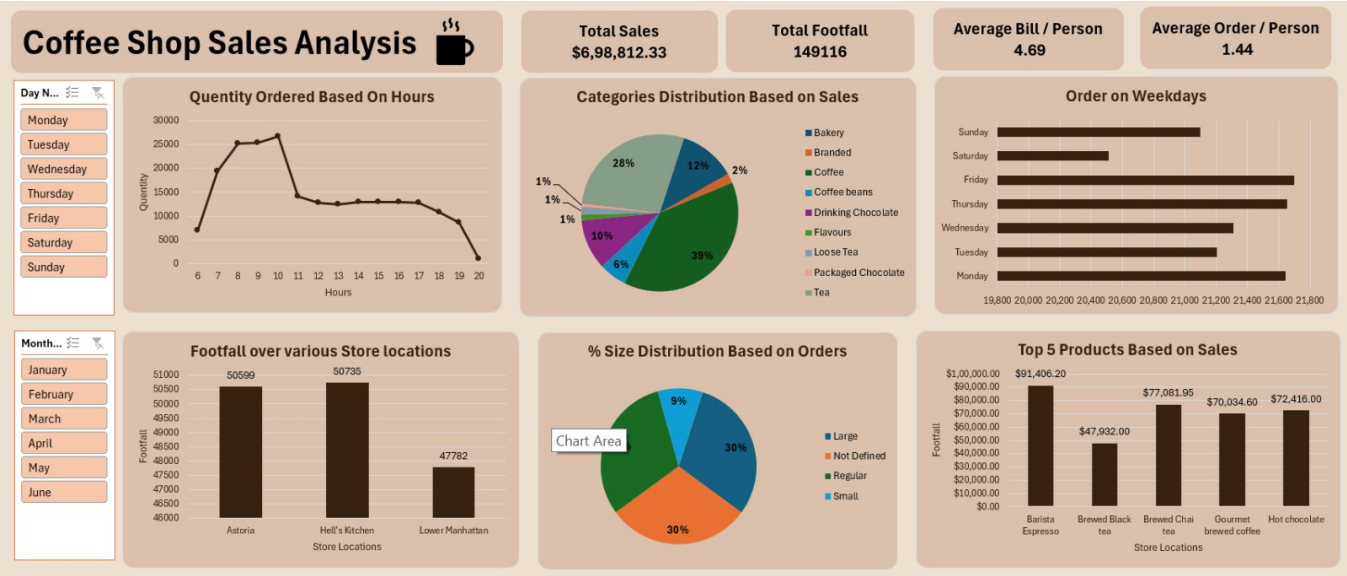
COFFEE SHOP SALES ANALYSIS

Excel-based Data Analytics Project Report

Project type	Primary tool	Records
Data Analytics	Microsoft Excel	149,116 transactions

Dashboard Preview



Dashboard screenshot supplied for this project. The placeholder image path from the Markdown README has been replaced with the actual screenshot.

1. Project Overview

Coffee Shop Sales Analysis is an Excel-based data analytics project that transforms raw transaction data into an interactive dashboard. It explores sales performance, transaction volume, product/category contribution, store activity, and hourly/weekday demand.

2. Business Problem

Coffee shop managers need a quick way to monitor revenue, demand patterns, product mix, and store-level activity. Reviewing thousands of individual transactions manually is time-consuming.

Analytical objective: summarize transaction data into clear metrics and visualizations so performance can be explored by date, weekday, month, hour, product category, product, size, and store location.

3. Dataset

The workbook contains 149,116 transaction records and fields including transaction ID, date and time, store ID/location, product ID/category/type/detail, quantity, unit price, and analysis helper fields.

Field	Meaning
transaction_id	Unique transaction identifier
transaction_date / transaction_time	Transaction date and time
store_id / store_location	Store identifier and location
product_id / product_category / product_type / product_detail	Product identifiers and product classification
transaction_qty	Quantity ordered
unit_price	Price per unit
Total_Bill	Transaction sales amount
Month Name / Month / Month Index	Month grouping and chronological sorting helpers
Day Name / Day of Week	Weekday grouping helpers
Hour	Hour extracted from transaction time
Custom	Size/variant field used in the workbook

Data note: Dashboard “Footfall” is based on transaction count. It should be interpreted as transaction volume, not a verified count of distinct visitors.

4. Tools & Excel Features

- Microsoft Excel
- Excel Tables for structured transaction data
- Calculated/helper columns for date and time analysis
- PivotTables for aggregation and comparison
- PivotCharts and Excel charts
- KPI calculations and number formatting
- Slicers for interactive filtering

5. End-to-End Project Workflow

1	Raw transaction dataset
↓	
2	Inspect fields and data types
↓	
3	Prepare Excel Table + helper fields
↓	
4	Build PivotTables
↓	
5	Calculate KPIs
↓	
6	Create charts + interactive dashboard
↓	
7	Validate totals
↓	
8	Document and publish on GitHub

Step 1 — Data inspection

Reviewed transaction fields, date/time columns, product attributes, store locations, quantities, and unit prices to understand what each record represents.

Step 2 — Data preparation

Structured transaction data in an Excel Table and used supporting fields for time-based analysis. Recommended checks include blanks, duplicate transaction IDs, inconsistent labels, and incorrect data types.

Step 3 — Feature engineering / helper columns

- Month Name / Month: group transactions by month.
- Day Name / Day of Week: compare weekday patterns.
- Hour: analyze demand by hour.
- Month Index: sort months chronologically rather than alphabetically.

Step 4 — PivotTable analysis

Aggregated sales, quantity, and transaction counts by weekday, month, hour, category, product, size, and store location.

Step 5 — KPI calculations

Summarized the main business metrics and placed them in KPI cards at the top of the dashboard.

Step 6 — Dashboard design

Combined KPI cards, line/bar/pie charts, and slicers into a single view. Day Name and Month slicers support interactive exploration.

Step 7 — Validation

Compared PivotTable grand totals with KPI totals and checked that visuals respond consistently to available filters.

6. Dashboard Walkthrough

KPI Cards

KPI	Dashboard value	Interpretation
Total Sales	$698,812.33	Sum of transaction bill amounts
Total Footfall	149,116	Transaction count used as footfall proxy
Average Bill / Person	$4.69	Total Sales ÷ transaction count
Average Order / Person	1.44	Total quantity ÷ transaction count

Metric definitions: Total Sales = SUM(Total_Bill); Transaction Count = COUNT(transaction_id); Average Bill / Transaction = Total Sales ÷ Transaction Count; Average Quantity / Transaction = SUM(transaction_qty) ÷ Transaction Count.

The dashboard labels say “per person,” but the source data does not establish unique customer identities. “Per transaction” is the more precise interpretation.

1. Quantity Ordered Based on Hours

A line chart showing total quantity ordered by hour. It helps identify busy periods and quieter hours for staffing and operational planning.

2. Category Distribution Based on Sales

A pie chart showing each product category’s share of total sales, helping compare category contribution and product mix.

3. Orders on Weekdays

A horizontal bar chart comparing transaction counts across weekdays to reveal differences in weekday demand.

4. Footfall over Store Locations

A column chart comparing transaction counts for Astoria, Hell’s Kitchen, and Lower Manhattan. This represents transaction volume, not unique visitors.

5. Size Distribution Based on Orders

A pie chart showing order distribution across Large, Not Defined, Regular, and Small size labels. “Not Defined” is retained as a category.

6. Top 5 Products Based on Sales

A column chart comparing the five displayed product types by sales amount. Displayed products include Barista Espresso, Brewed Chai tea, Hot chocolate, Gourmet brewed coffee, and Brewed Black tea.

Slicers

Day Name and Month slicers filter the connected PivotTables/charts. Depending on Excel report connections, a slicer may control only the PivotTables to which it is connected.

7. Key Observations

- Recorded sales total $698,812.33 across 149,116 transactions.
- June is the highest-sales month shown ($166,485.88), followed by May ($156,727.76).
- Coffee is the largest category in the category summary ($269,952.45), followed by Tea ($196,405.95).
- Hell’s Kitchen has the highest transaction count among the three stores (50,735), narrowly ahead of Astoria (50,599).
- Barista Espresso is the highest-sales item among the five displayed products ($91,406.20).

- Hourly quantity is strongest in the morning-to-midday range, with the highest plotted hour at 10:00.

These are descriptive findings from the workbook, not causal explanations. Operational decisions should also consider staffing, inventory, promotions, and other context.

8. Suggested GitHub Repository Structure

```
coffee-shop-sales-analysis/\n■■■ README.md\n■■■ data/\n■   ■■■ README.md\n■■■ images/\n■   ■■■ coffee-shop-dashb
```

If the dataset is large or redistribution is restricted, do not upload raw data. Include a data-source note instead. Share the workbook and screenshot only if you have permission.

9. How to Explore

- Clone or download the repository.
- Open the Excel workbook in desktop Microsoft Excel for full PivotTable/slicer functionality.
- Open the Dashboard sheet.
- Use Day Name and Month slicers to filter the dashboard.
- Review KPI cards and charts together to understand sales, volume, category mix, and store activity.

10. Skills Demonstrated

- Spreadsheet data analysis and preparation
- Date/time feature creation
- PivotTable aggregation
- KPI definition and calculation
- Data visualization and dashboard design
- Interactive filtering with slicers
- Business question framing and insight communication

11. Limitations & Future Improvements

- Transaction count is not equivalent to unique customer footfall.
- Add a data dictionary and documented cleaning log.
- Add a unique-customer field if the source supports customer-level analysis.
- Add targets, cost/profit data, and month-over-month comparisons if available.
- Rebuild in Power BI as a follow-up with a documented data model and measures.
- Add automated refresh and validation checks for repeatable reporting.

12. Author

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A portfolio learning project built to practice Excel-based data analysis and dashboard reporting.